

Erdős Problem #261

EP-261

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Are there infinitely many n such that there exists some $t \geq 2$ and distinct integers $a_1, \dots, a_t \geq 1$ such that $\frac{n}{2^n} = \sum_{1 \leq k \leq t} \frac{a_k}{2^{a_k}}$? Is this true for all n ? Is there a rational x such that $x = \sum_{k=1}^{\infty} \frac{a_k}{2^{a_k}}$ has at least $2^{\aleph_0}$ solutions?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>